Integrating Convolution and Sparse Coding for Learning Low-Dimensional Discriminative Image Representations

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***Abstract*— This work investigates the problem of efficiently learning discriminative low-dimensional (LD) representations of multiclass image objects. We propose a generic end-to-end approach that jointly optimizes sparse dictionary and convolutions for learning LOW-dimensional discriminative image representations, named *SparConvLow*, taking advantage of *c*onvolutional *n*eural *n*etworks (*CNN*s), dictionary learning, and orthogonal projections. The whole learning process can be summarized as follows. First, a *CNN* module is employed to extract high-dimensional (HD) preliminary convolutional features. Second, to avoid the high computational cost of direct sparse coding on HD *CNN* features, we learn sparse representation (SR) over a task-driven dictionary in the space with the feature being orthogonally projected. We then exploit the discriminative projection on SR. The whole learning process is consistently treated as an end-to-end joint optimization problem of trace quotient maximization. The cost function is well-defined on the product of the *CNN* parameters space, the *Stiefel* manifold, the *Oblique* manifold, and the *Grassmann* manifold. By using the explicit gradient delivery, the cost function is optimized via a geometrical stochastic gradient descent (SGD) algorithm along with the chain rule and the backpropagation. The experimental results show that the proposed method can achieve a highly competitive performance with the state-of-the-art (SOTA) image classification, object categorization, and face recognition methods, under both supervised and semi-supervised settings. The code is available at https://github.com/MVPR-Group/SparConvLow.**

***Index Terms*— Convolutional neural network (CNN), discriminative representation learning, geometric optimization, manifold, sparse coding.**

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I. Introduction

ONE core discovery in low-dimensional (LD) discriminative image representation learning is that the deep architecture of representations efficiently discovers the discrepancy underneath the observed image set. Appropriate LD representations of image data can significantly improve the performance of image discrimination learning algorithms [1], [2]. Once the model has learned discriminative features, it can perform well in various visual tasks, such as image classification [3], [4], [5], visualization [6] and clustering [6], [7], [8], face recognition [9], object detection, tracking, and categorization [3], [10], [11]. Specifically, the performance of such discrimination learning algorithms highly relies on the appropriate selection and organization of various purpose priors [1], such as sparsity [3], [12], [13], convolution [5], [14], [15], smoothness [1], linear subspace [16], [17], depth of explanatory factors [18], similarity relation [7], and graph and manifold [2], [19]. A suitable choice of the above knowledge priors has been found to be a convenient way to help the representation learner discover and disentangle some of the underlying discriminative factors hidden in image data [1], [2], [18]. Hence, the combination of knowledge priors is capable of facilitating visual discrimination learning tasks [1], [18]. This motivates us to develop an LD representation joint learning algorithm by taking advantage of several approximate knowledge priors, which can overcome the shortcomings of a specific single prior.

Among various knowledge priors, convolution, sparsity, and orthogonal projection for dimensional reduction (DR) via linear subspace or manifold have been verified as effective techniques for extracting discriminative features from images [1], [2], [12], [17], [19]. Convolutional neural networks (*CNN*s) have shown power on efficiently extracting local features of images [14], [20], [21]. The sparse coding methods represent images as sparse linear combinations of a shared dictionary, which becomes a dictionary learning (*DiL*) problem when the dictionary is trained to fit the input data. It is a well-known powerful tool to distill underlying nonlinear sparse structures of data that discover the discriminative information hidden in global domain [3], [10], [12], [13], [22], [23]. Orthogonal DR projections have the power to explore discriminative relations between high-dimensional (HD) data points and also have advantages in more aspects, such as

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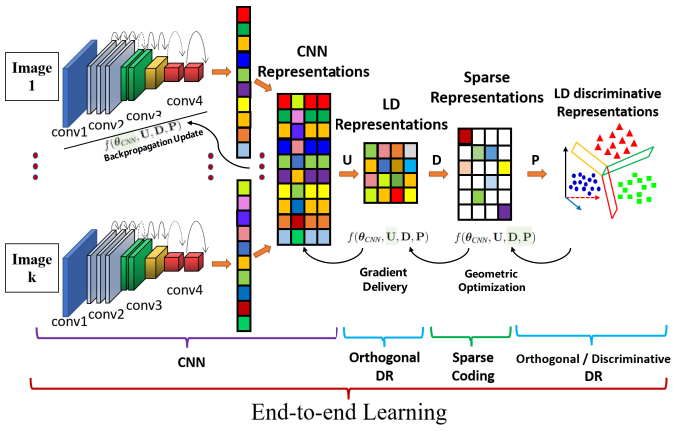


Fig. 1. Exemplar pipeline for classifying k images by using the proposed algorithm, i.e., *SparConvLow*. $\mathbf{U}, \mathbf{D}$, and $\mathbf{P}$ denote the orthogonal DR projection, the dictionary, and the final discriminative projection, respectively. Additionally, gradient backpropagation direction and updatable parameters are shown by right-to-left curved arrows.

better norm preservation to maintain gradient scalability when optimization, well-conditioned transformation, and information preservation in DR process [4], [24], [25].

In this study, in order to efficiently learn discrimination among HD images, we propose a novel joint learning paradigm by taking advantage of the *CNN*, the sparse coding, and the orthogonal DR projection. To facilitate description, we refer to the proposed learning scheme as *jointly optimizing SPARse dictionary and CONVolutions for learning LOW-dimensional discriminative image representations (SparConvLow)*. The motivation is to harness the untapped potential within the high-level semantic information of deep *CNN* features, particularly their sparse underlying structures. The *SparConvLow* aims to mine and exploit these crucial representations by applying DiL and dimension reduction, which can significantly improve the performance of image discrimination tasks. Moreover, the *SparConvLow* is designed to be naturally extended to semi-supervised learning environments, thereby enhancing the flexibility and learning ability of *CNN*s.

The proposed joint end-to-end learning scheme consists of three components, as shown in Fig. 1.

1) First, a *CNN* is conducted as an extractor of local features from raw images and generates fixed-length HD preliminary feature vectors.
2) Second, a traditional orthogonal DR projection is imposed on *CNN* feature vectors to obtain LD features. Such a DR reduces the time consumption of computing sparse coefficients from HD *CNN* feature vectors.
3) Third, a linear discriminative orthogonal projection is directly performed on sparse coefficients to learn the final LD discriminative representations.
4) Specifically, considering the image classification problem, popular classifiers, such as k-nearest neighbor (k-NN) or support vector machine (SVM), are applied to classify obtained LD representations.

In a nutshell, the overall learning system contains the *CNN* parameters, the orthogonal projection matrix on original *CNN* features, the dictionary matrix, and the discriminative projection matrix on sparse coefficients. The challenge is to train the proposed global learning scheme by using the end-to-end backpropagation algorithm. Thus, a gradient delivery strategy is then introduced to deliver the gradient layer by layer and achieve end-to-end training by using backpropagation. Furthermore, considering that the orthogonal projections and the dictionary admit the strict geometric structures, which can be expressed naturally by using matrix manifold [26]. In this study, we build a global cost function that models all parameters on the product space of the *CNN* parameters space, the *Stiefel* manifold, the *Oblique* manifold, and the *Grassmann* manifold. Then, we use a geometric gradient descent (GD) algorithm to smoothly optimize the global cost function. The main contributions of this work can be summarized into four aspects.

1) We propose *SparConvLow*, a novel approach that leverages the benefits of *CNN*, sparse coding, and orthogonal DR to enhance feature discrimination, offering improved data representation performance with robustness to spatial variance, and an efficient projection from HD to LD spaces.
2) We address the differentiability of sparse representations (SRs) with respect to (w.r.t.) a given dictionary and an orthogonal DR projection mapping matrix. Hence, we seamlessly integrate sparse coding into an end-to-end deep learning framework, and the solution of *SparConvLow* is treated as a problem of jointly learning parameters restricted to matrix manifolds.
3) By using the explicit gradient delivery, we present a mini-batch-based geometric stochastic gradient descent (SGD) algorithm, which can significantly reduce the dimension of optimization problems compared with the method that solves the problem in the ambient *Euclidean* space. Additionally, the well-defined parameters can improve the generalization of the proposed algorithm.
4) The learned orthogonal projections can be adapted to the sparse structure over *CNN* features, which improves the robustness against several cases, such as the small-size problem of training samples and the semi-supervised problem.

The rest of this article is organized as follows. In Section II, we review the related work. In Section III, we give the notations and preliminaries, including an introduction to sparse coding. In Section IV, we fully investigate the proposed *SparConvLow* framework, with both supervised and semi-supervised settings. A geometric SGD algorithm is developed in Section V. Section VI presents the implementation settings and numerical experiments. Finally, we conclude this work in Section VII.

II. Related Work

Efficiently exploring the discriminative relationships among HD image data points has been widely studied since its importance in many applications of computer vision (CV). Considering the HD image data points contain a high degree of irrelevant and redundant information which may greatly degrade the performance of some discrimination learning models, various LD representation learners are constructed to

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reduce such irrelevance and redundancy, in order to be able to store or transmit data in an efficient form.

A. Conventional Discriminative LD Representation Learning Methods

Conventional approaches represent HD data collection as the input lying in an LD smooth manifold [17], [27] or some underlying *subspaces* [28], called DR. Such a setting covers prominent examples such as linear discriminant analysis (LDA) [28], semi-supervised discriminant analysis (SDA) [29], Laplacian eigenmap (LE) [30], and orthogonal locality preserving projections (OLPP) [16], which encodes the HD data as a *dense undercomplete* representation over a dictionary with orthogonal columns [31], called orthogonal projections. Furthermore, by restricting orthogonal projections on appropriate matrix manifolds, e.g., *Stiefel* manifold [17], some geometric gradient methods can be convenient to solve the above DR problems, substituting for classical matrix decomposition [10], [26]. Hence, they are also well-known efficient tools for projecting multiclass HD data onto LD subspaces, by virtue of the low computational complexity and the low sample complexity [31]. In addition, such kind of orthogonal projection methods allow the classical DR to be integrated into larger learning architectures. However, they often suffer from noise corruption and the low efficiency for extracting nonlinear discrimination between images due to their shallow linear projection architecture.

Differently, by appealing to the prior of *sparsity*, SR is a well-known robust tool to explore the underlying nonlinear sparse structure of images [7], [11], [32]. The method of SR assumes that a signal $\wr \in \mathbb{R}^m$ can be sparsely represented by a linear combination of atoms in a dictionary $\mathbf{D} \in \mathbb{R}^{m\times q}$, $m \ll q$, i.e.,

$$\wr = \mathbf{D}\boldsymbol{\phi} + \boldsymbol{\varepsilon} \tag{1}$$

where $\boldsymbol{\phi} \in \mathbb{R}^q$ and $\boldsymbol{\varepsilon} \in \mathbb{R}^m$ denote the SR of $\mathbf{s}$ and the small additive residual, respectively. SR has achieved great success with wide applications in signal reconstruction, denoising, and inpainting [6], [32]. Moreover, this article focuses on exploring the discrimination of SRs or learning dictionaries for different tasks. Several works have empirically validated that it is easier to extract useful task-driven information in the sparse domain [22], [25]. Generally, in order to efficiently extract task-specific information, jointly learning both a dictionary and a task-specific transformation is widely adopted, leading to task-driven DiL approaches. Mairal et al. [12] presented a general formulation for supervised DiL adapted to both classification and regression tasks. Jiang et al. [33] proposed label-consistent k-singular value decomposition (LC-KSVD1/2) to learn a single overcomplete dictionary and an optimal linear classifier jointly. Quan et al. [34], [35] introduced a discriminative sparse coding approach that jointly learns a dictionary and an ensemble of classifiers instead of a single classifier for enhanced discrimination in complex data.

Although this joint learning strategy benefits the final task-specific predictors, it still suffers from heavy time complexity and enormous storage burden when the class number is large. Furthermore, it is difficult to learn extremely redundant dictionaries from HD inputs by computing (1), just depending upon regular computing resources [31]. Some works devised nonlinear DiL via kernel sparse coding to handle unknown LD manifolds of HD data, e.g., manifold-specific DiL [36] and Wasserstein DiL [37]. Most researchers turn to utilize the idea of depth network fitting iterative optimization algorithm or use the idea of a local computation to alleviate this problem, e.g., deep sparse coding network [38], [39] and convolutional DiL [22], [40].

Both DR and sparse DiL can also efficiently explore the discriminative relations among image data points that have already been preprocessed, such as the well-cropped tinny images/patches set with uniform scales, good alignment, and illumination. However, they may fail to solve the more general practical real-world CV problems. For example, they often suffer from the following two aspects.

1) Most raw photographic images have very high dimensions (millions of dimensions) under various changes, such as rotations, illuminations, centering offsets, and others.
2) The raw input images may contain multiple objects in a variety of scales, locations, and viewpoints with complex backgrounds.

B. Convolution Neural Networks

Typical representation learners for real-world images are appealing to the *CNN* [41], [42], which are based on the strong priors such as *convolution* and *pooling*, as well as *the depth* of the network [1]. The architecture of *CNN*s has been verified as an efficient tool to extract local features of images [14], [20], [21]. Outstanding even current records on visual tasks are commonly achieved by deep *CNN*s, such as VGGNet [41], Inception series [43], ResNet [14], and DenseNet [44]. It can be found that most deep *CNN*s for visual tasks are composed of dozens of convolutional layers, pooling layers, activation functions, and several fully connected (*FC*) layers associated with a label-driven regression. However, the convolution operation is generally considered a linear transformation, and operators that bring nonlinear transformation which is necessary are not learnable, e.g., activation and pooling. This strategy usually requires a large number of labeled training samples, resulting in prohibitive training efforts for a large number of classes. Furthermore, most existing *CNN*s mainly focus on learning representation only for a single image, and then recognizing objects separately. The accuracy of network training is highly dependent on the correct given image labels, rather than exploring the underlying relationship characteristics among image data points [14], [41], [44].

In order to enhance the discriminative capability of *CNN*s among HD data points, various methods are proposed by combining fashionable *CNN* models and the criteria that measure the relations among data points [2], [9], [45]. Similarly, more works combine the output of a deep *CNN* with DR, to find the LD discriminative representation of images, which are then treated as inputs to task-specific learning algorithms, such as image classifiers [9], [46], [47]. For

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example, FaceNet [9] uses triplets loss as the global cost function to find the LD face embedding in *CNN* features. Meanwhile, some have explored the combination of *CNN* and SR. However, such approaches are often constructed as separated two-stage encoding schemes [11], [48], i.e., the further sparse coding paradigm is independent of *CNN* feature extraction stage and the whole learning framework fails to achieve global optimization via a common end-to-end training algorithm. Recent developments in [49], [50], and [51] show that the convolutional layer, the *FC* layer, or the SoftMax layer of *CNN*s can be replaced by the *DiL* layer, with an end-to-end training scheme. However, there are still shortcomings and areas for improvement.

1) These methods often suffer from high computational complexity and have not fully leveraged the structured information hidden in *CNN* features, as well as with the application being limited to classification.
2) There is still a lack of research on the framework combining *CNN* with traditional DR methods, which may improve the discriminability of LD representations and the performance of downstream tasks. Meanwhile, such a joint framework has major issues such as global optimization that need to be overcome.

III. Notations and Preliminaries

This section first gives the notations and then introduces some technical details of sparse coding with elastic-net prior, which is the core component of the proposed *SparConvLow*. Finally, the sample complexity is provided to estimate the deviation amount of the empirical cost function from its expectation to the sparse coding problem. It plays a critical role in improving the computational efficiency of *SparConvLow*.

A. Notations

Vectors and matrices are denoted by bold lower case letters and upper case ones, respectively. $\boldsymbol{v}_i$ denotes the ith column of matrix $\boldsymbol{\mathcal{V}}$, $\boldsymbol{E}_{ij}$ denotes the matrix whose ith entry in the jth column equals one and all other entries are zero, $\mathrm{tr}(\cdot)$ denotes the trace of a matrix, $\mathcal{R}(\cdot)$ denotes the full vector of an SR, $\nabla f(\mathbf{x})$ and $(\partial f/\partial \mathbf{x})$ denote the gradient of f w.r.t. the vector $\mathbf{x}$ and the partial derivatives of f w.r.t. $\mathbf{x}$, respectively.

B. Sparse Coding

In order to exploit sparse factorization of (1), it is important to obtain a dictionary that makes the signal $\imath_i$ be accurately represented with a coefficient vector $\boldsymbol{\phi}_i$, which is as sparse as possible. Let $\boldsymbol{\mathcal{O}} := [\imath_1, \ldots, \imath_n] \in \mathbb{R}^{m\times n}$ denote the matrix that contains n independent identically distributed (i.i.d.) training samples as n columns, respectively. The purpose of *DiL* is to find the best dictionary for sparsely representing the elements of $\boldsymbol{\mathcal{O}}$. $\boldsymbol{\Phi} := [\boldsymbol{\phi}_1, \ldots, \boldsymbol{\phi}_n] \in \mathbb{R}^{q\times n}$ represents the matrix whose n columns correspond to n sparse transform coefficient vectors, respectively. Generally, classical *DiL* techniques can be summarized by the following optimization problem [32]:

$$\min_{\mathbf{D}\in\mathfrak{D}(m,q),\boldsymbol{\Phi}\in\mathbb{R}^{q\times n}} \sum_{i=1}^{n} \frac{1}{2}\|\imath_i - \mathbf{D}\boldsymbol{\phi}_i\|_2^2 + g(\boldsymbol{\phi}_i) \tag{2}$$

where $(1/2)\|\imath_i - \mathbf{D}\boldsymbol{\phi}_i\|_2^2$ denotes the reconstruction error caused by SRs, $g : \mathbb{R}^q \to \mathbb{R}^+$ denotes a function promoting the sparse structure of $\boldsymbol{\phi}$, and $\mathfrak{D}(m, q)$ denotes the predefined admissible solution set in $\mathbb{R}^{m\times q}$ for the dictionary $\mathbf{D}$.

Given an overcomplete dictionary $\mathbf{D} \in \mathfrak{D}(m, q)$ $(q > m)$, the SR problem in (1) can be solved by several ways. Suppose that the sparsity is measured by a combination of the ℓ_1-norm and the ℓ_2-norm, i.e.,

$$g(\boldsymbol{\phi}) := \lambda_1\|\boldsymbol{\phi}\|_1 + \lambda_2\|\boldsymbol{\phi}\|_2^2. \tag{3}$$

It corresponds to the elastic-net problem [52] that can be solved by

$$\boldsymbol{\phi}^* := \underset{\boldsymbol{\phi}\in\mathbb{R}^q}{\mathrm{argmin}} \frac{1}{2}\|\imath - \mathbf{D}\boldsymbol{\phi}\|_2^2 + \lambda_1\|\boldsymbol{\phi}\|_1 + \frac{\lambda_2}{2}\|\boldsymbol{\phi}\|_2^2. \tag{4}$$

Equation (4) provides a convenient way to generate the SR. $\lambda_1 \in \mathbb{R}^+$ and $\lambda_2 \in \mathbb{R}^+$ denote the regularization parameters, and they are selected to guarantee the stability and the uniqueness of the sparse solution, respectively.

In this study, the norm of each column $\mathbf{d}_i \in \mathbb{R}^m$ of $\mathbf{D}$ is constrained to be unit, i.e.,

$$\mathfrak{D}(m, q) := \left\{\mathbf{D} \in \mathbb{R}^{m\times q} \middle| \mathbf{d}_i^\top \mathbf{d}_i = 1\right\} \tag{5}$$

where $\mathfrak{D}(m, q)$ denotes a product of $q \times (m-1)$-dimension unit spheres. We also control the interaction between atoms by making $\mathbf{d}_i^{\mathrm{T}}\mathbf{d}_j = 0$ when $i \neq j$; thus, $\mathfrak{D}(m, q)$ corresponds to a $q(m-1)$-dimension smooth manifold, called *Oblique* manifold [26]. The constraint of the unit-norm column on the dictionary $\mathbf{D} \in \mathfrak{D}(m, q)$ is adopted to avoid the scale ambiguity of the dictionary.

IV. Methodology

In this section, we construct *SparConvLow*, i.e., an end-to-end discrimination learning framework that extends the sparse coding associated with orthogonal DR projections into a deep *CNN* structure. *SparConvLow* yields multiple levels of representation that can explore more intricate internal structures in HD image data (see [1]). Fig. 1 depicts the pipeline of *SparConvLow* model, which constructs two basic modules, i.e., the *CNN* module for feature extraction and the sparse dimension reduction module (*SparDR*) for learning discriminative image representations.

We first outline the data flow in the *SparConvLow* framework. Formally, let $\mathbf{Z} := [\mathbf{z}_1, \ldots, \mathbf{z}_n] \in \mathbb{R}^{m\times n}$ be a data collection. According to the architecture of *SparConvLow* (shown in Fig. 1), the convolutional features are obtained via an elementwise *CNN* mapping [14], [41], [42]

$$f_{\mathrm{CNN}} : \mathbf{z}_i \mapsto \mathbf{x}_i, \quad \text{for all } i = 1, \ldots, n \tag{6}$$

with $\mathbf{x}_i \in \mathbb{R}^m$ denoting the ith *CNN* feature vector. Hence, $\mathbf{x}_i$ is to feed the *SparDR* module, which forms a mapping

$$f_{\mathrm{SparDR}} : \mathbf{x}_i \mapsto \mathbf{y}_i, \quad \text{for all } i = 1, \ldots, n \tag{7}$$

with $\mathbf{y}_i \in \mathbb{R}^d$ being an LD representation vector.

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TABLE I
DIMENSION (m) OF SOME EXEMPLAR *CNN* FEATURES THAT DENOTE THE COLUMN VECTORS BEFORE THE FLATTENING LAYER

Dimension	Raw image	AlexNet-5 [21]	ZF-5 [20]	SPP-Net [42]	DenseNet-121 [44]	VGG-13/16 [41]	ResNet-18/34 [14]
m	224×224 $= 50176$	$6 \times 6 \times 256$ $= 9216$	$6 \times 6 \times 256$ $= 9216$	21×256 $= 5376$	$7 \times 7 \times 384$ $= 18816$	$7 \times 7 \times 512$ $= 25088$	$7 \times 7 \times 512$ $= 25088$

The combination of the *CNN* module and the *SparDR* module forms *SparConvLow* learning system, and its global mapping could be written by

$$f : \mathbf{z}_i \mapsto \mathbf{x}_i \mapsto \mathbf{y}_i, \quad \text{for all } i = 1, \ldots, n. \tag{8}$$

Since the *CNN* module has been already well-established, in the following, the *SparDR* module is first introduced in Section IV-A, which is constructed as a three-layer joint representation learning module that leverages both aforementioned sparse coding and orthogonal DR projections. The whole *SparConvLow* learning framework with an end-to-end training scheme is then presented in Section IV-B. Several practical examples are finally shown in Section IV-C.

A. SparDR Module

In this section, we construct the *SparDR* module to learn LD discriminative representations from *CNN* features. Since the dimensions of *CNN* features are extremely high (as shown in Table I), in order to reduce the sample complexity of sparse coding, we first construct the sparse coding problem in the orthogonal projected space of *CNN* features. Then, we present the global cost function for the *SparDR* module, which adopts the construction of sparse coding with elastic-net in the framework of trace quotient maximization, to extract LD discriminative representations of SRs of input data.

1) DiL in Dimension-Reduced Space: Given a vector $\mathbf{x} \in \mathbb{R}^m$, it could denote a raw image vector with high resolution, or a *CNN* feature vector [41], [42]. It is intuitively suggested that learning a dictionary usually requires an extremely high demand for computing resources [31], according to Table I and the sample complexity analysis of *DiL*. Note that, for better readability, the mathematical analysis of the sample complexity of *DiL* is shown in *Section I of the Supplementary Materials*. It is worth noting that the sample complexity increases significantly with an increase in the dictionary size. For example, the corresponding *sample complexity* of sparse coding on VGG-16 features [41] is much larger than 6×10^8, i.e., m, q with $q > m$ and $m = 25\,088$. Furthermore, the very high dimensionality associated with limited labeled training samples is usually influenced by the *small sample size problem*. Therefore, it is necessary to conduct an additional learning paradigm to reduce the computing sizes of m and q before the sparse coding, which motivates the following joint learning algorithm.

In order to reduce the computational complexity of DiL, a commonly used method is to conduct a linear DR projection on $\mathbf{x}_i$, i.e.,

$$\hat{\mathbf{x}}_i = \mathbf{U}^\top \mathbf{x}_i \tag{9}$$

where $\mathbf{U} \in \mathbb{R}^{m \times l}, \hat{\mathbf{x}}_i \in \mathbb{R}^l, l \ll m$. Thus, the sparse factorization of $\hat{\mathbf{x}}_i$ w.r.t. a dictionary $\mathbf{D} \in \mathfrak{D}(l, k)$ can be computed by

$$\hat{\mathbf{x}}_i = \mathbf{D}\boldsymbol{\phi}_i + \boldsymbol{\varepsilon} \tag{10}$$

where $i = 1, \ldots, n$.

With the aim of achieving the computational efficiency, many classic linear DR methods restrict the row vectors in $\mathbf{U}$ to be orthonormal, i.e., $\mathbf{U}$ belongs to the *Stiefel* manifold

$$\mathrm{St}(l, m) := \{\mathbf{U} \in \mathbb{R}^{m \times l} | \mathbf{U}^\top \mathbf{U} = \mathbf{I}_l\}. \tag{11}$$

By solving the elastic-net problem on dimension-reduced features $\hat{\mathbf{x}}$ over $\mathbf{D}$, we rewrite (4) as follows:

$$\boldsymbol{\phi}^*_{\mathbf{x}}(\mathbf{U}, \mathbf{D}) := \underset{\boldsymbol{\phi} \in \mathbb{R}^k}{\operatorname{argmin}} \frac{1}{2}\|\mathbf{U}^\top \mathbf{x} - \mathbf{D}\boldsymbol{\phi}\|_2^2 + \lambda_1 \|\boldsymbol{\phi}\|_1 + \frac{\lambda_2}{2}\|\boldsymbol{\phi}\|_2^2 \tag{12}$$

with $\lambda_1 \in \mathbb{R}^+$ and $\lambda_2 \in \mathbb{R}^+$.

For the sparse factorization problem represented by (12), all solutions share the fact that the SR w.r.t. a specific dictionary is unique. Thus, each sparse vector $\boldsymbol{\phi}^*_{\mathbf{x}}(\mathbf{U}, \mathbf{D})$ can be treated as an implicit function in $\mathbf{x}$, $\mathbf{U}$, and $\mathbf{D}$. Under appropriate assumptions, the sparse solution $\boldsymbol{\phi}^*_{\mathbf{x}}(\mathbf{U}, \mathbf{D})$, over variables $\mathbf{x}$, $\mathbf{U}$, and $\mathbf{D}$, has closed-form expressions. Furthermore, the implicit function $\boldsymbol{\phi}^*_{\mathbf{x}}(\mathbf{U}, \mathbf{D})$ is locally differentiable w.r.t. $\mathbf{x}$, $\mathbf{U}$, and $\mathbf{D}$ and their first-order derivatives of $\boldsymbol{\phi}^*_{\mathbf{x}}(\mathbf{U}, \mathbf{D})$ w.r.t. $\mathbf{x}$, $\mathbf{U}$, and $\mathbf{D}$ have also closed-form expressions, respectively. Details will be discussed in Section V and *the Supplementary Materials*.

With SRs of all samples and their first-order derivatives being calculated, specific task-driven learning algorithms can be directly applied to these sparse coefficients to extract further representations. This allows us to integrate the developed sparse coding problem associated with its various downstream tasks into a deep learning architecture.

2) Unified Trace Quotient Framework for Learning Discriminative Features on SRs: As suggested by the works of [7], [10], [12], and [22], SRs can be further processed to reveal the underlying discriminative information, which benefits supervised learning tasks particularly. The key assumption of this work is that the sparse presentations carry more discriminative information than the original data space. By following the trace quotient maximization framework, in this section, we design a cost function that jointly learns both the dictionary and the two-layer orthogonal transformations, to efficiently extract discriminant features on SRs.

Let $\mathbf{X} := [\mathbf{x}_1, \ldots, \mathbf{x}_n] \in \mathbb{R}^{m \times n}$ be a collection of n data points in $\mathbb{R}^m$. Recalling the data projection achieved by (9), let $\hat{\mathbf{X}} = [\hat{\mathbf{x}}_1, \ldots, \hat{\mathbf{x}}_n] \in \mathbb{R}^{l \times n}$ denote the projected samples in $\mathbb{R}^l$. The corresponding sparse coefficients are denoted by $\boldsymbol{\Phi} =$

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$[\boldsymbol{\phi}_1, \ldots, \boldsymbol{\phi}_n]^\top \in \mathbb{R}^{n\times k}$, which are confined to be learned from the projected samples set $\hat{\mathbf{X}}$, by solving the sparse regression problem in (12).

One way for learning discriminative projections for (7) is performing a DR from sparse coefficients. Formally, given sparse coefficients $\boldsymbol{\phi}_i \in \mathbb{R}^k$, the *SparDR* module aims to find a representation $\mathbf{y}_i \in \mathbb{R}^d$ with a lower dimensionality from $\boldsymbol{\phi}_i \in \mathbb{R}^k$ with $d < k$, via a mapping $\mu\colon \mathbb{R}^k \to \mathbb{R}^d$, which captures certain application-dependent properties of the sparse coefficients. Specifically, in this study, we confine ourselves to the form of orthogonal projections as $\mu(\boldsymbol{\phi}_i) := \mathbf{V}^\top \boldsymbol{\phi}_i$ with $\mathbf{V} \in \mathrm{St}(k, d)$.

One generic algorithmic framework to find optimal $\mathbf{V} \in \mathrm{St}(k, d)$ is formulated as a maximization problem of the so-called *trace quotient* or *trace ratio* [53], i.e.,

$$\underset{\mathbf{V}\in\mathrm{St}(k,d)}{\operatorname{argmax}} f_{\mathrm{SparDR}}(\mathbf{V})$$
$$f_{\mathrm{SparDR}}(\mathbf{V}) := \frac{\mathrm{tr}(\mathbf{V}^\top \boldsymbol{\Phi}^\top \mathbf{A} \boldsymbol{\Phi} \mathbf{V})}{\mathrm{tr}(\mathbf{V}^\top \boldsymbol{\Phi}^\top \mathbf{B} \boldsymbol{\Phi} \mathbf{V})} \tag{13}$$

where $\boldsymbol{\Phi}^\top \mathbf{A} \boldsymbol{\Phi} \in \mathbb{R}^{n\times n}$ and $\boldsymbol{\Phi}^\top \mathbf{B} \boldsymbol{\Phi} \in \mathbb{R}^{n\times n}$ are constructed to measure the "similarity" between sparse coefficient pairs $(\boldsymbol{\phi}_i, \boldsymbol{\phi}_j), \forall i, j = n$ according to the specific problems. They are assumed to be symmetric and positive definite (SPD) for simplicity [17]. In this work, they are represented as smooth functions to measure the discrepancy in $\boldsymbol{\Phi}$. Since they have different structures under different projection methods and constraints, the methodological details and examples will be given and discussed in Section IV-C.

Recalling that the solutions to the trace quotient cost function in (13) are restricted on *Stiefel* manifold; thus, the function in (13) is invariant w.r.t. orthonormal basis changes of $\mathbf{V}$, i.e.,

$$f_{\mathrm{SparDR}}(\mathbf{V}^*\boldsymbol{\Theta}) = f_{\mathrm{SparDR}}(\mathbf{V}^*), \quad \text{for all } \boldsymbol{\Theta}\boldsymbol{\Theta}^\top = \mathbf{I}_d. \tag{14}$$

If $\mathbf{V}^* \in \mathrm{St}(k, d)$ is a solution of the maximization problem (13), then, its all rotations $\mathbf{V}\boldsymbol{\Theta}$ for any $\boldsymbol{\Theta} \in \mathrm{St}(d, d)$ are also solutions to (13), i.e., solutions of the maximization problem in (13) are rotation-invariant.

The property in (14) indicates that the solution set of the problem in (13) is the set of all d-dimensional linear subspaces in $\mathbb{R}^k$, which induces functions defined on the *Grassmann* manifold $\mathrm{Gr}(d, k)$. In this study, $\mathrm{Gr}(d, k)$ is defined by the set of all k-dimensional linear subspaces of $\mathbb{R}^k$. Formally, it is defined as

$$\mathrm{Gr}(d, k) := \left\{ \mathbf{V}\mathbf{V}^\top | \mathbf{V} \in \mathrm{St}(k, d) \right\}. \tag{15}$$

Let $\mathbf{P} = \mathbf{V}\mathbf{V}^\top$. Thus, by restricting $\mathbf{P} \in \mathrm{Gr}(d, k)$, the second-order problem in (13) can be represented as a first-order trace quotient maximization problem on *Grassmann* manifold

$$f_{\mathrm{SparDR}}(\mathbf{P}) := \frac{\mathrm{tr}\left(\boldsymbol{\Phi}\mathbf{A}\boldsymbol{\Phi}^\top\mathbf{P}\right)}{\mathrm{tr}\left(\boldsymbol{\Phi}\mathbf{B}\boldsymbol{\Phi}^\top\mathbf{P}\right)}. \tag{16}$$

The transformation from (13) to (16) makes the uniqueness of the solution more convenient.

It is worth noticing that when $\mathbf{P}$ is given, $\mathbf{V}$ can be obtained by the QR-factorization

$$\mathbf{P} = \mathbf{V}\mathbf{R} \tag{17}$$

where $\mathbf{V} \in \mathrm{St}(k, d)$ and $\mathbf{R}$ is the invertible upper triangular with nonnegative entries on the diagonal (see [54]).

Overall, by combining the aforementioned operators, the mapping for f_{SparDR} in (7) is implemented by

$$f_{\mathrm{SparDR}}\colon \mathbf{x}_i \overset{\mathbf{U}}{\mapsto} \hat{\mathbf{x}}_i \overset{\mathbf{D}}{\mapsto} \boldsymbol{\phi}_i \overset{\mathbf{V}}{\mapsto} \mathbf{y}_i, \quad \text{for all } i = 1, \ldots, n \tag{18}$$

which involves the parameters $\mathbf{U}, \mathbf{D}$, and $\mathbf{V}$. Recalling the sparse regression problem in (12), let $\boldsymbol{\Phi}_{\mathbf{X}}(\mathbf{U}, \mathbf{D}) = [\boldsymbol{\phi}^*_{\mathbf{x}_1}(\mathbf{U}, \mathbf{D}), \ldots, \boldsymbol{\phi}^*_{\mathbf{x}_n}(\mathbf{U}, \mathbf{D})] \in \mathbb{R}^{k\times n}$ denote SRs of $\hat{\mathbf{X}}$ w.r.t. $\mathbf{U}$ and $\mathbf{D}$. By treating the $\boldsymbol{\Phi}_{\mathbf{X}}(\mathbf{U}, \mathbf{D})$ as implicit function in $\mathbf{X}$, $\mathbf{U}$, and $\mathbf{D}$, we rewrite the problem defined in (16) with all aforementioned ingredients as maximizing the following cost function:

$$f_{\mathrm{SparDR}}\colon \mathrm{St}(l, m) \times \mathfrak{D}(l, k) \times \mathrm{Gr}(d, k) \to \mathbb{R}$$
$$f_{\mathrm{SparDR}}(\mathbf{U}, \mathbf{D}, \mathbf{P}) := \frac{\mathrm{tr}\left(\boldsymbol{\Phi}\mathbf{A}\boldsymbol{\Phi}^\top\mathbf{P}\right)}{\mathrm{tr}\left(\boldsymbol{\Phi}\mathbf{B}\boldsymbol{\Phi}^\top\mathbf{P} + \sigma\mathbf{I}_k\right)} \tag{19}$$

with $\boldsymbol{\Phi} := \boldsymbol{\Phi}_{\mathbf{X}}(\mathbf{U}, \mathbf{D})$, where f_{SparDR} denotes a first-order smooth function defined on a product manifold $\mathrm{St}(l, m) \times \mathfrak{D}(l, k) \times \mathrm{Gr}(d, k)$. $\sigma > 0$ in (19) guarantees that the denominator is greater than zero always holds. We call the module defined in (19) as *SparDR*.

B. *End-to-End* SparConvLow *Learning System*

With the *SparDR* module at hand, in this section, we follow the pipeline depicted in Fig. 1 and construct the *SparConvLow* framework. The former extracts local features from raw images to generate a fixed-length HD preliminary feature vector, and the latter is in charge of learning discriminative representations from *CNN* features.

Formally, recalling the mappings defined in (6)–(8), *SparConvLow* system is composed of f_{CNN} in (6) and f_{SparDR} in (7) and (19). f_{CNN} involves all remaining *CNN* parameters, denoted by $\boldsymbol{\theta}_{\mathrm{CNN}}$, and f_{SparDR} involves a triple of matrices, i.e., $\mathbf{U}, \mathbf{D}$, and $\mathbf{P}$. Hereinafter, we represent the complete *SparConvLow* parameters by

$$\boldsymbol{\theta}_{\mathrm{SparConvLow}} = \{\boldsymbol{\theta}_{\mathrm{CNN}}, \mathbf{U}, \mathbf{D}, \mathbf{P}\}.$$

Let us further assume the bridge between f_{CNN} and f_{SparDR} is smooth, i.e., the gradient of f_{SparDR} could be delivered to f_{CNN}, we have

$$\nabla f\left(\boldsymbol{\theta}^L_{\mathrm{CNN}}\right) = \nabla f\left(\mathbf{X}\right) \prod_{L}^{L+1} \left(\frac{\partial f^L_{\mathrm{CNN}}\left(\boldsymbol{\theta}^L_{\mathrm{CNN}}, \mathbf{z}^L\right)}{\partial \boldsymbol{\theta}^L_{\mathrm{CNN}}} \right) \mathbf{h}_{\boldsymbol{\theta}^L_{\mathrm{CNN}}} \tag{20}$$

where $(\partial f^L_{\mathrm{CNN}}(\boldsymbol{\theta}^L_{\mathrm{CNN}}, \mathbf{z}^L)/\partial \boldsymbol{\theta}^L_{\mathrm{CNN}})\mathbf{h}_{\boldsymbol{\theta}^L_{\mathrm{CNN}}}$ can be conveniently computed by the standard *CNN*. Note that, to alleviate the understanding, in *Section II of the Supplementary Materials*, we provide a comprehensive description of the gradient's backpropagation delivery process from the *SparDR* module to the *CNN* component. Additionally, proofs of differentiability of f_{SparDR} and first-order derivation $\nabla f(\mathbf{X})$ are also given.

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The derivatives of *SparDR* module have been transferred to the *CNN* module layer by layer. Optimizing *SparConvLow* system could be written as a global cost function by

$$f : \mathcal{C} \times \mathrm{St}(l,m) \times \mathfrak{D}(l,k) \times \mathrm{Gr}(d,k) \to \mathbb{R}$$
$$f(\boldsymbol{\theta}_{\mathrm{CNN}}, \mathbf{U}, \mathbf{D}, \mathbf{P}) := \frac{\mathrm{tr}\,(\boldsymbol{\mathcal{A}}\mathbf{P})}{\mathrm{tr}\,(\boldsymbol{\mathcal{B}}\mathbf{P} + \sigma \mathbf{I}_k)} \tag{21}$$

where $\mathcal{C}$ denotes the feasible solution set of $\boldsymbol{\theta}_{\mathrm{CNN}}$ and is commonly identified by *Euclidean* space. $\boldsymbol{\mathcal{A}}$ and $\boldsymbol{\mathcal{B}}$ are according to the formats of (19), and they can be rewritten by

$$\boldsymbol{\mathcal{A}} = \boldsymbol{\Phi}_{f_{\mathrm{CNN}}(\boldsymbol{\theta}_{\mathrm{CNN}}, \mathbf{Z})}(\mathbf{U}, \mathbf{D}) \mathbf{A} \boldsymbol{\Phi}^{\top}_{f_{\mathrm{CNN}}(\boldsymbol{\theta}_{\mathrm{CNN}}, \mathbf{Z})}(\mathbf{U}, \mathbf{D}) \tag{22}$$
$$\boldsymbol{\mathcal{B}} = \boldsymbol{\Phi}_{f_{\mathrm{CNN}}(\boldsymbol{\theta}_{\mathrm{CNN}} \mathbf{Z})}(\mathbf{U}, \mathbf{D}) \mathbf{B} \boldsymbol{\Phi}^{\top}_{f_{\mathrm{CNN}}(\boldsymbol{\theta}_{\mathrm{CNN}}, \mathbf{Z})}(\mathbf{U}, \mathbf{D}) \tag{23}$$

with f_{CNN} being an elementwise *CNN* mapping in (6). Therein, $\mathbf{x}_i = f_{\mathrm{CNN}}(\boldsymbol{\theta}_{\mathrm{CNN}}, \mathbf{z}_i)$ for a given signal $\mathbf{z}_i$, and $\boldsymbol{\Phi}_{f_{\mathrm{CNN}}(\boldsymbol{\theta}_{\mathrm{CNN}}, \mathbf{Z})}(\mathbf{U}, \mathbf{D})$ denotes SRs of *CNN* features $\{\mathbf{x}_i\}$. Hereinafter, we name (21) as *SparConvLow* function. If f is smooth w.r.t. $\boldsymbol{\theta}_{\mathrm{CNN}}, \mathbf{U}, \mathbf{D}$, and $\mathbf{P}$, by regarding (21) as the global main cost function and using the chain rule, the backpropagation algorithm can be used for end-to-end training, which is investigated in Section V.

C. Examples for SparConvLow: LDA and SDA

The construction of *SparConvLow* function is determined by various structured functions for $\mathbf{A}$ and $\mathbf{B}$ in (19) and (21). It covers supervised, semi-supervised, and unsupervised DR methods [55], [56], such as LDA [28], SDA [29], and LE [30]. In the following, we utilize LDA and SDA as examples to evaluate the proposed problem (21). It is important to note that this problem is a generic joint disentangling framework; hence, it is not limited to such methods presented here. More examples are presented in *Section III of the Supplementary Materials*.

The goal of LDA and SDA [28], [29] is to find an LD representation of the HD data, so that the interclass scatter is maximized, while the within-class scatter is minimized. Formally, given data with c classes, including $n_{\mathcal{L}}$ labeled samples and $n_{\mathcal{U}}$ unlabeled samples. We can define the intraclass scatter matrix $\mathbf{A}^{\mathcal{L}}$ as $\sum_{i=1}^{n_{\mathcal{L}}}(1/n_i)\mathbf{e}_i\mathbf{e}_i^{\top} - (1/n_{\mathcal{L}})\mathbf{1}_{n_{\mathcal{L}}}\mathbf{1}_{n_{\mathcal{L}}}^{\top}$, where $\mathbf{1}_{n_{\mathcal{L}}} = [1, \ldots, 1]^{\top} \in \mathbb{R}^{n_{\mathcal{L}}}$, $\mathbf{e}_i$ is an $n_{\mathcal{L}}$-dimensional vector with $\mathbf{e}_i(\iota) = 1$ if $i = c_\iota$; 0 otherwise. Herein, c_ι is the class label of ιth sample, $\iota = 1, 2, \ldots, n_{\mathcal{L}}$. The within-class scatter matrix $\mathbf{B}^{\mathcal{L}}$ can be written as $\mathbf{I}_{n_{\mathcal{L}}} - (1/c)\sum_{i=1}^{c}\mathbf{e}_i\mathbf{e}_i^{\top}$. In the semi-supervised setting, matrices $\mathbf{A}$ and $\mathbf{B}$ of SDA are composed of $\mathbf{A}^{\mathcal{L}}$, $\mathbf{B}^{\mathcal{L}}$, and graph Laplacian $\mathbf{L}$ (defined in *Section III of the Supplementary Materials*) with $\beta \in \mathbb{R}$, which are shown as

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}^{\mathcal{L}} & \mathbf{0}_{n_L \times n_u} \\ \mathbf{0}_{n_u \times n_L} & \mathbf{0}_{n_u \times n_u} \end{bmatrix}$$
$$\mathbf{B} = \begin{bmatrix} \mathbf{B}^{\mathcal{L}} & \mathbf{0}_{n_L \times n_u} \\ \mathbf{0}_{n_u \times n_L} & \mathbf{0}_{n_u \times n_u} \end{bmatrix} + \beta \mathbf{L}.$$

V. Optimization Algorithm for *SparConvLow*

In this section, we develop a batch-based geometric GD algorithm to train the proposed *SparConvLow*. Note that in the following, $\boldsymbol{\Phi}_{\mathbf{X}}(\mathbf{U}, \mathbf{D})$ is abbreviated as $\boldsymbol{\Phi}$.

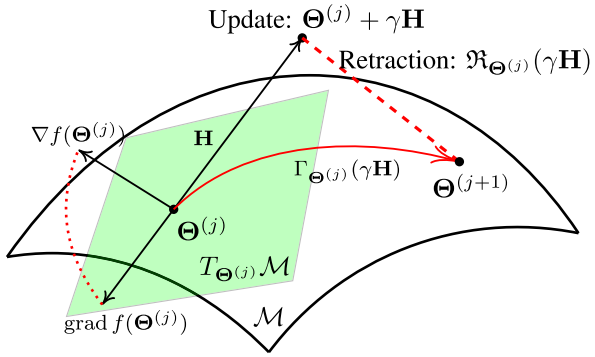


Fig. 2. Some essential concepts of the geometric optimization algorithm. First, it depicts two nearby points $\boldsymbol{\Theta}^{(j)}$ and $\boldsymbol{\Theta}^{(j+1)}$ on a manifold $\mathcal{M}$ together with the tangent space at $\boldsymbol{\Theta}^{(j)}$ (green area). The tangent space, $T_{\boldsymbol{\Theta}^{(j)}}\mathcal{M}$, is a real vector space consisting of all tangent vectors that pass through $\boldsymbol{\Theta}^{(j)}$. Second, it shows an update from the point $\boldsymbol{\Theta}^{(j)}$ to the point $\boldsymbol{\Theta}^{(j+1)}$ in a search direction $\mathbf{H} \in T_{\boldsymbol{\Theta}^{(j)}}\mathcal{M}$ along the curve $\Gamma_{\boldsymbol{\Theta}^{(j)}}(\gamma\mathbf{H})$. The geodesic $\Gamma_{\boldsymbol{\Theta}^{(j)}}(\gamma\mathbf{H})$ in the direction of $\mathbf{H}$ connects the two points $\boldsymbol{\Theta}^{(j)}$ and $\boldsymbol{\Theta}^{(j+1)}$, with γ being the step size. Third, it describes that how to approximate the geodesic $\Gamma_{\boldsymbol{\Theta}^{(j)}}(\gamma\mathbf{H})$ by using the retraction $\mathfrak{R}_{\boldsymbol{\Theta}^{(j)}}(\gamma\mathbf{H})$. Finally, it introduces the gradient at $\boldsymbol{\Theta}^{(j)}$, i.e., the *Euclidean* gradient $\nabla f(\boldsymbol{\Theta}^{(j)})$ and the Riemannian gradient $\mathrm{grad}\, f(\boldsymbol{\Theta}^{(j)}) \in T_{\boldsymbol{\Theta}^{(j)}}\mathcal{M}$. The latter is the orthogonal projection of the former onto the tangent space at $\boldsymbol{\Theta}^{(j)}$.

A. Geometric SGD Algorithm

By using the layerwise training strategy of the backpropagation algorithm, the updates of $\boldsymbol{\theta}_{\mathrm{CNN}}$ and $\mathbf{U}, \mathbf{D}$, and $\mathbf{P}$ can be separable, i.e., at first, $\mathbf{U}, \mathbf{D}$, and $\mathbf{P}$ are updated by maximizing (21), then $\boldsymbol{\theta}_{\mathrm{CNN}}$ are updated layer by layer via the chain rule.

In what follows, we first consider the optimization of $\mathbf{U}, \mathbf{D}$, and $\mathbf{P}$. Recalling that $\mathbf{U} \in \mathrm{St}(l,m), \mathbf{D} \in \mathfrak{D}(l,k)$, and $\mathbf{P} \in \mathrm{Gr}(d,k)$, $\mathbf{U}, \mathbf{D}$, and $\mathbf{P}$ are admitting approximate geometric manifold structures. Directly updating $\mathbf{U}, \mathbf{D}$, and $\mathbf{P}$ in *Euclidean* space, i.e., one takes steps proportional to the negative of the *Euclidean* gradient of the function, would change geometric manifold structures of them. The geometric GD algorithm is necessary for solving such a problem [26].

The general solution to the optimization problem [see (21)] on the matrix manifold can be denoted by $\boldsymbol{\Theta} \in \boldsymbol{\mathcal{M}}$, where $\boldsymbol{\Theta}$ is an element of $\boldsymbol{\mathcal{M}}$. Here, $\boldsymbol{\mathcal{M}}$ denotes an example of $\mathrm{St}(l,m)$, $\mathfrak{D}(l,k)$, or $\mathrm{Gr}(d,k)$. Fig. 2 illustrates some concepts on $\boldsymbol{\mathcal{M}}$ to facilitate understanding. The detailed overview of the optimization on matrix manifolds can be found in [26]. Before the introduction of geometric GD algorithm on $\boldsymbol{\mathcal{M}}$, in the following, we first compute the Riemannian gradients of f w.r.t. $\mathbf{U}$, $\mathbf{D}$, and $\mathbf{P}$. Note that, to alleviate the cognitive burden imposed on readers by intricate derivations in the main text, the derivations of the Euclidean gradient $\nabla f(\mathbf{U})$, $\nabla f(\mathbf{D})$, and $\nabla f(\mathbf{D})$ are detailed in *Section II of the Supplementary Material*. The latter part of this section is primarily dedicated to demonstrating how to transform these Euclidean gradients into Riemannian gradients through manifold retraction operations and to perform unconstrained manifold optimization, which empirically exhibits better convergence properties.

As shown in Fig. 2, the Riemannian gradient of f at $\boldsymbol{\Theta} \in \boldsymbol{\mathcal{M}}$ is an element of tangent space $T_{\boldsymbol{\Theta}}\boldsymbol{\mathcal{M}}$. The direction of the Riemannian gradient of $f(\boldsymbol{\Theta})$ is the steepest-ascent

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direction f at $\boldsymbol{\Theta}$. Let $T_{\mathbf{U}}\mathrm{St}(l,m)$, $T_{\mathbf{D}}\mathfrak{D}(l,k)$, and $T_{\mathbf{P}}\mathrm{Gr}(d,k)$ be the tangent spaces of $\mathbf{U} \in \mathrm{St}(l,m)$, $\mathbf{D} \in \mathfrak{D}(l,k)$, and $\mathbf{P} \in \mathrm{Gr}(d,k)$, respectively. As introduced in Fig. 2, we now compute the individual gradients at $\mathbf{U}$, $\mathbf{D}$ and $\mathbf{P}$. Having computed the Euclidean gradients of f w.r.t. $\mathbf{U}$, $\mathbf{D}$, and $\mathbf{P}$, the Riemannian gradients at $\mathbf{U}$, $\mathbf{D}$, $\mathbf{P}$, denoted by $\mathrm{grad}(\mathbf{U})$, $\mathrm{grad}(\mathbf{D})$ and $\mathrm{grad}(\mathbf{P})$, are computed as follows:

$$\begin{aligned}\mathrm{grad}(\mathbf{U}) &= \boldsymbol{\tau}_{\mathbf{U}}\big(\nabla f(\mathbf{U})\big)\\ \mathrm{grad}(\mathbf{D}) &= \boldsymbol{\Pi}_{\mathbf{D}}\big(\nabla f(\mathbf{D})\big)\\ \mathrm{grad}(\mathbf{P}) &= \boldsymbol{\pi}_{\mathbf{P}}\big(\nabla f(\mathbf{P})\big)\end{aligned} \tag{24}$$

where $\boldsymbol{\tau}_{\mathbf{U}}(\mathbf{M}_1)$, $\boldsymbol{\Pi}_{\mathbf{D}}(\mathbf{M}_2)$, and $\boldsymbol{\pi}_{\mathbf{P}}(\mathbf{M}_3)$ are the orthogonal projections of some arbitrary points $\mathbf{M}_1 \in \mathbb{R}^{m\times l}$, $\mathbf{M}_2 \in \mathbb{R}^{l\times k}$, and $\mathbf{M}_3 \in \mathbb{R}^{k\times k}$ onto the tangent spaces $T_{\mathbf{U}}\mathrm{St}(l,m)$, $T_{\mathbf{D}}\mathfrak{D}(l,k)$, and $T_{\mathbf{P}}\mathrm{Gr}(d,k)$, respectively. Then, we compute the Riemannian gradients of f w.r.t. the parameters $\mathbf{U}$, $\mathbf{D}$, and $\mathbf{P}$ as

$$\begin{aligned}\boldsymbol{\tau}_{\mathbf{U}}\big(\nabla f(\mathbf{U})\big) &= \nabla f(\mathbf{U}) - \mathbf{U}(\nabla f(\mathbf{U}))^{\top}\mathbf{U}\\ \boldsymbol{\Pi}_{\mathbf{D}}\big(\nabla f(\mathbf{D})\big) &= \nabla f(\mathbf{D}) - \mathbf{D}\,\mathrm{diag}(\mathbf{D}^{\top}\nabla f(\mathbf{D}))\\ \boldsymbol{\pi}_{\mathbf{P}}\big(\nabla f(\mathbf{P})\big) &= [\mathbf{P},[\mathbf{P},\nabla f(\mathbf{P})]]\end{aligned} \tag{25}$$

where the matrix commutator "bracket" is defined by $[\mathbf{A},\mathbf{B}] := \mathbf{AB} - \mathbf{BA}$, and $\mathrm{ddiag}(\mathbf{M})$ denotes a diagonal matrix whose diagonal entries correspond to those of $\mathbf{M}$. The above formulas regarding the geometry of $\mathrm{St}(l,m) \times \mathfrak{D}(l,k) \times \mathrm{Gr}(d,k)$ are derived in [26].

Algorithm 1 *SGD-SparConvLow* Algorithm

Input : Given training set $\mathbf{X} \in \mathbb{R}^{m\times n}$, which is divided into η batches and each batch with the size of τ, i.e., $\mathbf{X} := [\mathbf{X}_1,\ldots,\mathbf{X}_\eta] \in \mathbb{R}^{m\times\eta\tau}$, regularization parameters λ_1, λ_2, T and γ ;

Output: $(\boldsymbol{\theta}^*_{CNN}, \mathbf{U}^*, \mathbf{D}^*, \mathbf{P}^*) \in \mathcal{C} \times St(l,m) \times \mathfrak{D}(l,k) \times Gr(d,k)$;

Step 1: Compute initializations for $\boldsymbol{\theta}^0_{CNN}$, $\mathbf{U}^{(0)}$, $\mathbf{D}^{(0)}$, $\mathbf{P}^{(0)}$, and initialize $j = -1$;

Step 2: Set $j = j + 1$;

Step 3: Compute sparse coefficients $\boldsymbol{\Phi}_{\mathbf{X}_i}(\mathbf{U}^{(j)},\mathbf{D}^{(j)})$ for each $\boldsymbol{\phi}_{\mathbf{x}}(\mathbf{U}^{(j)},\mathbf{D}^{(j)})$ via Eq. (12), with random $i = 1,\cdots,\eta$;

Step 4: Compute $\mathbf{H}^{(j+1)}_{\mathbf{U}} = -\mathrm{grad}(\mathbf{U}^{(j+1)})$, $\mathbf{h}^{(j+1)}_{\mathbf{d}} = -\mathrm{grad}(\mathbf{D}^{(j+1)})$, $\mathbf{H}^{(j+1)}_{\mathbf{P}} = -\mathrm{grad}(\mathbf{P}^{(j+1)})$ via Eq. (24) ;

Step 5: Update $(\mathbf{U}^{(j+1)},\mathbf{D}^{(j+1)},\mathbf{P}^{(j+1)}) \leftarrow$ project $(\mathbf{U}^{(j)},\mathbf{D}^{(j)},\mathbf{P}^{(j)}) + \gamma\mathbf{H}^{(j)}$ onto $St(l,m)$, $\mathfrak{D}(l,k)$, $Gr(d,k)$, by computing Retractions defined in Eq. (26) ;

Step 6: Update $\boldsymbol{\theta}^{j+1}_{CNN}$ layer by layer according to Eq. (20) ;

Step 7: If $\|f\|$ is small enough, or $j \geq T$, stop. Otherwise, go to Step 2 ;

Finally, by assembling the Riemannian gradients, some geometric gradient algorithms can be easily employed to solve the maximization problem (21). We use the classical heuristic in SGD algorithms [12] to train our proposed *SparConvLow*.

In each iteration of the training algorithm, appropriate mini-batch examples are adopted. Algorithm 1 briefly summarizes an SGD algorithm for maximizing the function f [see (21)]. In Algorithm 1, the key is how to update the point $\boldsymbol{\Theta}^{(j)}$ to the point $\boldsymbol{\Theta}^{(j+1)}$ along the commonly used *geodesic*, denoted by the red curve $\Gamma_{\boldsymbol{\Theta}^{(j)}}(\gamma\mathbf{H})$ in Fig. 2. The computation of *geodesic* is often more computationally demanding (see [26]). In this study, a simplified approach, called retraction, is adopted instead of *geodesic*.

A retraction $\mathfrak{R}_{\boldsymbol{\Theta}}\colon T_{\boldsymbol{\Theta}}\mathcal{M} \to \mathcal{M}$ is a smooth mapping from the tangent space $T_{\boldsymbol{\Theta}}\mathcal{M}$ to the manifold, such that the evaluation $\mathfrak{R}_{\boldsymbol{\Theta}}(0) = \boldsymbol{\Theta}$ and the derivative $\mathcal{D}\Gamma_{\boldsymbol{\Theta}}(0)\colon T_{\boldsymbol{\Theta}}\mathcal{M} \to T_{\boldsymbol{\Theta}}\mathcal{M}$ is the identity mapping. Formally, let us define $\mathbf{H}_{\mathbf{U}}$, $\mathbf{H}_{\mathbf{D}}$, and $\mathbf{H}_{\mathbf{P}}$ the search directions at $\mathbf{U}$, $\mathbf{D}$, and $\mathbf{P}$, and the retractions defined on matrix manifolds $\mathrm{St}(l,m)$, $\mathfrak{D}(l,k)$, and $\mathrm{Gr}(d,k)$ are

$$\begin{cases}\mathfrak{R}_{\mathbf{V}}\gamma\mathbf{H}_{\mathbf{U}}) := \big(\mathbf{V} + \gamma\mathbf{H}_{\mathbf{U}}\big)_Q\\ \mathfrak{R}_{\mathbf{D}}(\gamma\mathbf{H}_{\mathbf{D}}) := \big[\mathfrak{R}_{\mathbf{d}_1}(\gamma\mathbf{h}_{\mathbf{d}_1}),\ldots,\mathfrak{R}_{\mathbf{d}_k}(\gamma\mathbf{h}_{\mathbf{d}_k})\big]\\ \mathfrak{R}_{\mathbf{P}}(\gamma\mathbf{H}_{\mathbf{P}}) := \zeta_{\mathbf{P}}(\gamma\mathbf{H}_{\mathbf{P}})\mathbf{P}\zeta_{\mathbf{P}}(\gamma\mathbf{H}_{\mathbf{P}})^{\top}\end{cases} \tag{26}$$

with

$$\begin{aligned}\mathfrak{R}_{\mathbf{d}}(\gamma\mathbf{h}_{\mathbf{d}}) &= \frac{\mathbf{d}+\gamma\mathbf{h}_{\mathbf{d}}}{\|\mathbf{d}+\gamma\mathbf{h}_{\mathbf{d}}\|_2}\\ \zeta_{\mathbf{P}}(\gamma\mathbf{H}_{\mathbf{P}}) &:= \big(\boldsymbol{I}_k + \gamma(\mathbf{H}_{\mathbf{P}}\mathbf{P} - \mathbf{P}\mathbf{H}_{\mathbf{P}})\big)_Q\end{aligned}$$

where $\mathbf{H}_{\mathbf{D}} = [\mathbf{h}_{\mathbf{d}_1},\ldots,\mathbf{h}_{\mathbf{d}_k}]$, and $(\cdot)_Q$ denotes the unique QR-factorization of an invertible matrix with the positive diagonal entries in its upper triangular part. $\boldsymbol{I}_k$ denotes the $k \times k$ identity matrix, whose diagonal entries in the upper triangular part are positive. An intuitive description for retractions is illustrated in Fig. 2. Once the Riemannian gradients are obtained, the way to update the point is via Step 4 to Step 5 in Algorithm 1. Furthermore, by treating *CNN* features as a hub for transferring gradients, the parameters $\boldsymbol{\theta}_{\mathrm{SparConvLow}}$ of *SparConvLow* could be easily updated in Step 6. Initializations for $\boldsymbol{\theta}_{\mathrm{SparConvLow}}$ are shown in *Section IV of the Supplementary Material.*

VI. Numerical Experiments

In this section, we first discuss the parameter settings in experiments and then demonstrate our experimental results. To facilitate description, the framework that combines the proposed *SparDR* module with preliminary *CNN*, e.g., ResNet-18 [14], is referred to as *ResNet-18-SparConvLow*. Additionally, as examples described in Section IV-C and *Section III of the Supplementary Material*, we can combine the *SparDR* module with LDA, SDA, and LE, denoted by SparLDA, SparSDA, and SparLE, respectively.

A. Experimental Settings

Throughout all following experiments, σ in (21) is set to $\sigma = 10^{-3}$ by experience. n, n_{train}, n_{test}, and $n_{\mathrm{train}D}$ represent the number of all images with c classes, the number of training

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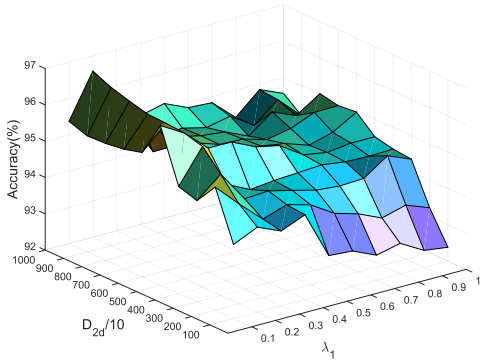


Fig. 3. Impact of the parameters λ_1, D_{2d}, and n_{train} on the *Accuracy* of the CIFAR-10 dataset.

samples, the number of testing samples, and the number of samples used for training the dictionary, respectively. Intuitively, $\mathbf{D}_{2d} = k$ represents the number of columns in a dictionary. Hereafter, $n = n_{\text{train}} + n_{\text{test}}$, $n_{\text{train}D} \leq n_{\text{train}}$, and $\mathbf{D}_{2d} \leq n_{\text{train}D}$.

In this study, we utilize different mainstream deep *CNN* models combine with the *SparDR* module, e.g., PreActResNet18 [57], EffNet(b0) [58], VGG-13/16 [41], and ResNet-18/34 [14]. As described in Algorithm 1, *SparConvLow* is based on a mini-batch training strategy. Batch size τ can be adjusted depending on computational and learning efficiency. Most of the experiments in this study are set to 64 or 256. The learning rate for *CNN*s starts from 0.01 in conjunction with a momentum term of 0.9. Let m, l, and d denote the dimensions of *CNN* features representation space, the dimension-reduced data subspace, and the final targeted LD discriminative representation subspace, respectively. m is often determined by the *CNN* structure (see Table I). For LDA/SDA-based *SparConvLow*, the d is set to $c-1$ as suggested [28]. The d in LE-based *SparConvLow* is set by practice. To speed up the computation, $l = 256$ for tiny image datasets, and $l = c - 1$ for large datasets. $\lambda_2 \in (0, (\lambda_1/10))$ [52] leads to the emphasis on the sparsity of SRs, and we set λ_2 to 10^{-7}.

The selection of parameters $\lambda_1, n_{\text{train}D}$, and $\mathbf{D}_{2d}$ and their impact on the performance of *SparConvLow* are also investigated, which can be seen in *Section V of the Supplementary Material*. Several conclusions are summarized as follows. Generally, the accuracy increases with $\mathbf{D}_{2d}$, $n_{\text{train}D}$, and the ratio of $n_{\text{train}D}$ to $\mathbf{D}_{2d}$. As λ_1 increases, the sparsity of SRs becomes higher, while the accuracy rises first and then decreases. Therefore, it is important to find an appropriate λ_1 to ensure the tradeoff between sparsity and performance. Furthermore, we take the experiment on the CIFAR-10 [59] dataset as an example, and Fig. 3 shows the classification accuracy by setting different λ_1 and $\mathbf{D}_{2d}$. Note that, one of the abscissa ($D_{2d}/10$) refers to the number of training samples of each category, with the range from 100 to 1000. $n_{\text{train}D} = \mathbf{D}_{2d}$, and λ_1 is selected from [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1]. The highest accuracy on the CIFAR-10 appears when λ_1 is set to 0.2, and both $n_{\text{train}D}$ and $\mathbf{D}_{2d}$ are 10 000.

In the following, we verify the effectiveness of our proposed *SparConvLow* on both tiny image datasets and large-scale image datasets, as well as the face datasets (shown in *Section VI of the Supplementary Material*).

TABLE II
TOP-1 ACCURACY (%) ON THE CIFAR-10 DATASET

CNN module	Accuracy (%)		
	Network	+ SVM / k-NN	+ SparLDA/ SparLE
ResNet-18 [14]	95.23	96.20/ 95.77	**97.19**/ 96.55
ResNet-34 [14]	95.69	95.90/ 94.18	**96.57**/ 96.16
VGG-13 [41]	88.18	88.08/ 87.77	**91.31**/ 90.23
VGG-16 [41]	88.88	88.82/ 88.80	**93.73**/ 90.13
PreActResNet18 [57]	91.14	91.82/ 91.68	**95.45**/ 92.66
EffNet(b0) [58]	91.67	93.03/ 92.83	**93.60**/ 92.96

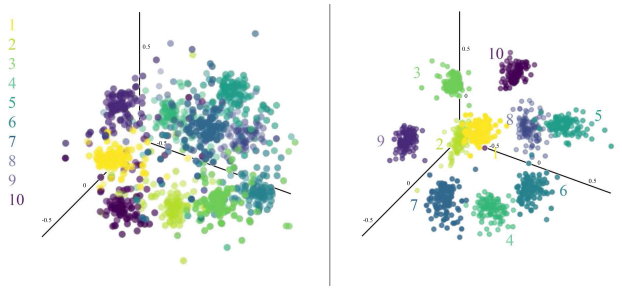


Fig. 4. Embedding visualization is realized by t-SNE: 3-D t-SNE visualization result of *CNN* features (left) and the visualization result of LD representation obtained by the proposed framework (right).

B. Classification on Datasets With Tiny Images

The performance of our method is first validated by experiments on tiny image datasets, such as the CIFAR-10 dataset [59], the SVHN dataset [60], and the Fashion-MNIST dataset [61]. As small-scale image datasets, they can quickly and effectively test the proposed algorithm or model. We standardized the images according to the standard value and mean value of the images in the training set.

1) Classification on the CIFAR-10 Dataset: Table II compares *SparConvLow* with several state-of-the-art (SOTA) *CNN*s, including ResNet-18/34 [14], VGG-13/16 [41], PreActResNet18 [57], and EffNet(b0) [58]. Then, the *CNN* followed by SVM or k-NN, denoted by +SVM/k-NN, is also used to verify the effectiveness of *SparConvLow*. It can be seen from Table II that the performance of the conventional classifier (+SVM/k-NN) is the same as or even lower than that of its backbone *CNN*. By integrating the proposed SparLDA/SparLE and the backbone *CNN*, all the classification results have been significantly improved.

In order to explore the ability of *SparConvLow* to obtain discriminant image representations, Fig. 4 intuitively compares the 3-D t-distributed stochastic neighbor embedding (t-SNE) visualization of *CNN* features with final LD representations of *SparConvLow*. As depicted in Fig. 4, the 3-D representations of *CNN* features shown on the left are hardly possible to be

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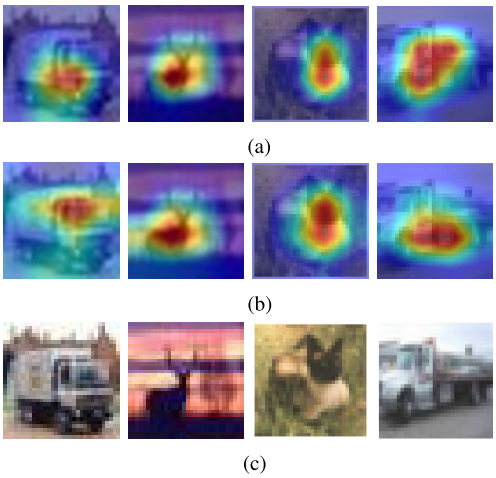


Fig. 5. Visual interpretation of ResNet-18 on the CIFAR-10 dataset. (a) Heatmaps generated from the pretraining *CNN* (ResNet-18). (b) Heatmaps generated from *SparConvLow* with ResNet-18. (c) Raw images.

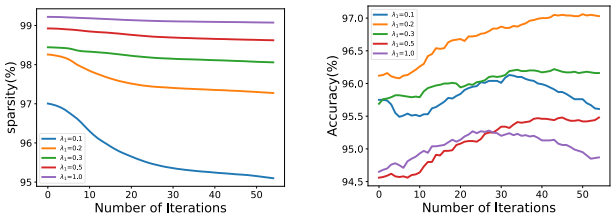


Fig. 6. Impact of λ_1 on the sparsity and on the accuracy of *SparConvLow* by using *ResNet-18-SparConvLow* on the CIFAR-10 dataset. Note that the abscissa is the number of training iterations, and the batch size is 64. (a) ResNet-18 λ_1-sparsity. (b) ResNet-18 λ_1-accuracy.

clustered or grouped. In particular, boundaries between each pair of images are completely entangled. It is evident that visualization powered by *SparConvLow*, i.e., the right one in Fig. 4, leads to direct clustering of the images. In short, *SparConvLow* highly improves the ability of discrimination learning, compared to raw *CNN*s.

Additionally, locating the class-discriminative regions in the image is a common practice to evaluate a *CNN*-based classification method [62]. Fig. 5 shows the comparison of the size change and movement of the model's attention before and after the addition of the *SparDR* module. The model's attention could be observed via the heatmaps in Fig. 5, in which the main focus area becomes larger, the boundary seems to be more blurred, and the color transitions become smoother.

Fig. 6 shows the impact of constraint factor λ_1 on the classification accuracy and the sparsity. In general, with the increasing number of iterations, the sparsity of SRs gradually decreases, but the accuracy is improved. Fig. 6(a) and (b) shows that the larger the constraint factor λ_1 is, the greater the sparsity is, but the improvement in accuracy diminishes. The same results go for *ResNet-34-SparConvLow* and the VGGNet in the proposed *SparConvLow* framework. When we set $\lambda_1 = 0.2$ and $D_{2d} = n_{\text{train}D} = 1000$, and train *ResNet-18-SparConvLow* about 50 iterations, it achieves the highest accuracy of 97.17%.

In Table III, we show the semi-supervised classification results of the proposed SparSDA and some SOTA semi-supervised methods including MixMatch [63], ReMixMatch [64], and FixMatch [65]. The results show that the model can classify unlabeled samples with higher accuracy because of the clustering information learned by *SparDR* module, with more details being in *Section VI of the Supplementary Material*.

2) Classification on the SVHN Dataset and Fashion-MNIST Dataset: On the SVHN dataset [60], we mainly compare the performance of the proposed framework with *CNN* methods. The results on the SVHN dataset are shown in Table IV. The classification accuracy of the proposed *SparConvLow* is 1%,

TABLE III
SEMI-SUPERVISED CLASSIFICATION ERROR RATE (%) ON THE CIFAR-10 DATASET

Method	80 Lables	250 Lables	500 Lables	1,000 Lables
MixMatch [63]	28.94	10.92	8.96	7.56
ReMixMatch [64]	-	6.27	-	5.73
FixMatch [65]	9.77	4.84	4.58	4.50
(20 Epochs) MixMatch	54.00	35.43	29.35	22.68
(20 Epochs) MixMatch + SparSDA	**52.36**	**33.92**	**28.04**	**21.98**
MixMatch + SparSDA	27.70	10.42	8.06	-
FixMatch + SparSDA	**4.68**	**4.50**	**4.42**	**4.39**

TABLE IV
TOP-1 ACCURACY (%) ON THE SVHN DATASET

CNN module	Accuracy (%)	
	Network	+ *SparLDA*
ResNet-18 [14]	95.54	**96.53**
ResNet-34 [14]	94.96	**97.32**
PreActResNet18 [57]	94.17	**96.90**
VGG-13 [41]	94.03	**96.94**
VGG-16 [41]	94.56	**97.21**
VGG-19 [41]	93.10	**94.52**

TABLE V
CLASSIFICATION ACCURACY (%) ON THE FASHION-MNIST DATASET

Method	Accuracy
ResNet-18 [14]	97.17
ResNet-34 [14]	97.55
LC-KSVD2 [33]	79.27
D-KSVD [66]	78.38
ADDL [67]	82.10
SRC [13]	79.86
CDPL-Net [51]	90.69
ResNet-18-SparConvLow	**99.25**
ResNet-34-SparConvLow	**99.26**

TABLE VI
CLASSIFICATION ERROR RATE (%) ON THE IMAGENET-1K DATASET

CNN module	*Top-1 test Error Rate* (%)	
	Network -15 Epochs / -Full-Trained	+ *SparDR* -15 Epochs / -Full-Trained
ResNet-18	50.57 / 30.24	44.33 / 29.96
ResNet-34	47.59 / 26.70	33.94 / 25.72

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TABLE VII
CLASSIFICATION ACCURACY (%) ON THE CALTECH-101 AND CALTECH-256 DATASETS

Method	Caltech101			Caltech256		
	$n_{\text{train}} = 5$	$n_{\text{train}} = 15$	$n_{\text{train}} = 30$	$n_{\text{train}} = 15$	$n_{\text{train}} = 30$	$n_{\text{train}} = 60$
ResNet-18 [14]	78.22	89.11	91.58	70.20	74.92	84.67
ResNet-34 [14]	80.69	87.62	92.57	73.67	74.22	83.84
K-SVD [32]	49.80	65.20	73.20	25.33	30.67	-
SRC [13]	48.80	64.90	70.70	27.86	33.33	-
D-KSVD [66]	49.60	65.10	73.00	27.79	33.72	48.88
LC-KSVD2 [33]	54.00	67.70	73.60	28.90	34.32	-
LDA-Sparlow [10]	56.44	73.82	76.86	38.05	43.27	55.74
DDLCN-3 [71]	60.21	74.62	81.98	47.65	54.28	62.42
ResNet-18-SparConvLow	**81.19**	90.26	92.90	71.61	**78.60**	88.66
ResNet-34-SparConvLow	80.69	**90.76**	**93.23**	**76.60**	77.73	**88.67**

2.36%, and 2.73% higher than the corresponding backbone networks ResNet-18, ResNet-34, and PreActResNet18, respectively. When *CNN* modules are VGG-13/16/19, the average accuracy is increased by 2.36%.

On the Fashion-MNIST dataset [61], we compare *SparConvLow* with outstanding DiL/SR methods, including traditional *DiL* methods, deep *DiL* methods, e.g., LC-KSVD2 (LC-KSVD with a joint classification error and label consistency regularization term) [33], D-KSVD [66], ADDL [67], SRC [13], and CDPL-Net [51]. In Table V, we set the parameters of traditional methods as the same as those in [51], with $D_{2d} = 2000$. *ResNet-18-SparConvLow* and *ResNet-34-SparConvLow* improve the original network accuracy by 2.08% and 1.71%, respectively. These results are also much higher than reported DiL/SR methods [33], [51], [66], [67].

Overall, the above results reveal that the *SparDR* module associated with the end-to-end training mechanism of *SparConvLow* exactly improves the performance of the backbone *CNN*s and has the ability to extract better discriminative image representations.

C. Object Categorization on Large-Scale Image Datasets

The experiments above have shown that the proposed *SparConvLow* had excellent performance on small-scale image classification. In the following, we investigate its object categorization performance on large-scale datasets, including the ImageNet-1K dataset [21], the Caltech-101 dataset [68], and the Caltech-256 dataset [69]. The subset of the ImageNet dataset [70] employed in the *ILSVRC* competition is widely used to evaluate the performance of *CNN*s. Learning on the Caltech-101 dataset is often considered to be difficult since the number of images in each category is uneven. However, the Caltech-256 dataset even contains objects with a great variety of poses and changeable locations within the images. The existence of background clutter and occlusion results in higher intraclass diversity, which makes the classification much more difficult. Considering the lack of training samples, we explore the generalization ability of feature extraction layers (i.e., the convolutional layers pretrained on the ImageNet) to Caltech datasets. This is the well-known transferring capability of convolutional layers, which has been verified by various classical *CNN*s [20], [41].

1) ImageNet Classification: On the *ILSVRC-2012* dataset, we follow the same practices of previous works [20], [21], [41]. The images are preprocessed with a 224 × 224 crop that is picked from the center or the four corners of the entire image, and standardized according to the mean and variance of the training set. Models are trained for 15 epochs and 1000 epochs (full-trained) on four 2080Ti GPUs with a batch size of 256, and the learning rate starts from 0.001 in conjunction with a momentum term of 0.9. The classification performance is evaluated by using the *Top-1 test error rate*. All results are shown in Table VI.

The *Top-1 test error rates* of ResNet-18 and ResNet-34 are 50.24% and 46.131%, respectively. *ResNet-18 SparConvLow* and *ResNet-34 SparConvLow* reduce the *Top-1 test error rates* by 5.91% and 12.19%, respectively. In addition, in the *ResNet-34 SparConvLow* framework, the sparsity of the SR of images is as high as 97.08%, which indicates that these SRs can be embedded into a much lower subspace.

2) Object Categorization on the Caltech Dataset: We load *CNN*s (e.g., ResNet-34) which have been pretrained on the ImageNet dataset, and fine-tune the model on the Caltech-101 and Caltech-256 datasets, respectively. We further initialize the dictionary for the *SparDR* module and train the whole *SparConvLow* framework. We randomly select 15 training samples in each category for initializing and training the dictionary, i.e., $n_{\text{train}D} = 15\,000$. We also set the second dimension of dictionary D_{2d} to be equal to $n_{\text{train}D}$. For both the Caltech-101 dataset and the Caltech-256 dataset, we follow the common experimental setups in [42] and then generate several random splits into training and testing data. In addition, on the Caltech-101 dataset, we randomly generate three splits, each of which contains 5, 15, and 30 training images per class. For each category of the Caltech-256 dataset, 15, 30, or 60 images are randomly sampled for training. Except for training samples, the rest of the samples in the dataset are used for testing.

Table VII shows the object categorization accuracy of *SparConvLow*, in comparison with the backbone *CNN* approaches and *DiL* methods. As can be seen from Table VII, the ImageNet-pretrained raw *CNN* approaches bring at least 13% improvement over *DiL* approaches. The proposed *ResNet-18-SparConvLow* and *ResNet-34-SparConvLow* achieve even higher. The main reason for raw *CNN* achieving better

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results is that deep neural networks could explore complex discriminative information that is hidden in imagery data. Furthermore, the trace quotient criteria-based *SparConvLow* globally improves the capability of discrimination learning. Hence, the proposed *SparConvLow* still performs best in scenarios such as large-scale datasets with only a few training samples.

VII. Conclusion

In this study, we presented an end-to-end discrimination learning approach, called *SparConvLow*, to find LD discriminative representations of multiclass imagery objects. The parameters of the *SparConvLow* cost function are well-modeled as a product of the *CNN* parameters space, the *Stiefel* manifold, the *Oblique* manifold, and the *Grassmann* manifold. By using the differentiability of solutions of sparse regression, *SparConvLow* could be well-adapted to the SOTA *CNN* methods and ensembles a differentiable deep learning framework. Hence, by using the explicit gradient delivery, *SparConvLow* could be efficiently solved by a proposed Riemannian SGD algorithm. *SparConvLow* takes advantage of *CNN*, sparse DiL, and orthogonal projections. The experimental results show that the proposed approach has a stronger competitive performance in comparison with SOTA methods. Moreover, the proposed *SparConvLow* is flexible and can be extended to deeper *CNN* models and more complex scenes, i.e., applying in more general cases of DR, even unsupervised learning problems with orthogonality constraints, zero-shot problem, etc. Having overcome the differentiability challenge of sparse coding, we can seamlessly integrate it into modern deep learning pipelines. The DiL community can further explore sparse priors with various models for interpretable deep feature learning. Considering the manifold optimization techniques that enhance optimization efficiency, the potential of such techniques and their extended applications across various tasks deserve attention. Another possible future direction is to extend to multimodal contrastive learning and investigate how the proposed framework can enhance representation discrimination and generalization across modal relationships.

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